Outliers & Cleaning Data

Exam code: 9MA0

Download PDF

Written by: Amber
Updated on 25 November 2025

SaveMyExams

Did this video help you? Yes No

Outliers

What are outliers?

- Outliers are extreme data values that do not fit with the general pattern of the data
- They can come from one or two extreme events or from mistakes in the data collection
- Outliers will affect some statistics that are calculated from the data
  - They can have a big effect on the mean, but not on the median or usually the mode
  - The range will be completely changed by a single outlier, but the interquartile range will not be affected
  - When calculating the mean or the range it is important to decide whether the outlier(s) should be included in the calculations
    - The question will tell you whether to include the outliers or not
    - You may have to decide which value is the outlier to be removed
    - In general outliers are included if they are a valid piece of data and excluded if it is likely that they are erroneous

How are outliers calculated?

- Most of the time within this syllabus the outliers will be a particular distance either side of the **interquartile range**

Help me understand

Stuck on something?

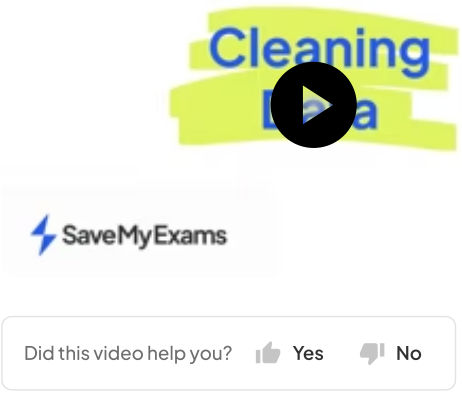
Ask questions and get help with exactly what you're revising.

Got it

        - A value that is greater than $Q_3 + k$ (interquartile range)
        - k is a constant that will be given to you in the exam, commonly k=1.5
- Outliers could also be situated a number of **standard deviations** away from the **mean**
    - The most common way to calculate an outlier will be using the formulae
        - A value that is less than $\bar{x} - k\sigma$
        - A value that is greater than $\bar{x} + k\sigma$
        - k is a constant that will be given to you in the exam, commonly $k = 2$

How are outliers represented on box plots?

- On a box plot an outlier is represented as a cross either side of the maximum or minimum value
- If the maximum or minimum value is discovered to be an outlier, the **new** maximum or minimum value will need to be found for the box plot
    - If the data value just above the minimum or just below the maximum is known, this will become the new value
    - If the data value is not known, the new minimum or maximum will become the outlier boundary



Cleaning data

When should data be cleaned?

- The cause of the outlier should be examined by looking into the context of the data
- For example:
    - a test score of over 100% would most likely be a data collection error
    - a single salary that is much higher than the others would likely be for the CEO of the company

cleaning the data

- It is important to consider very carefully whether you should remove the data value or not
  - If the data value is not an error it should not be removed from the data

- If a data value is removed from the data set before calculations are carried out, a justification for the removal of the outlier must be made
- Cleaning data also involves removing missing data and errors

Worked Example

The ages, in years, of a number of children attending a birthday party are given below:

2, 7, 5, 4, 8, 4, 6, 5, 5, 29, 2, 5, 13,

An outlier is defined as an observation that falls more than $1.5 \times$ the interquartile range above the upper quartile or below the lower quartile

(i) Identify any outliers within the data set.

(ii) Clean the data by deciding which values should be removed, justify your answer.

Answer:

(i) Start by putting the data in order of size

2, 2, 4, 4, 5, 5, 5, 5, 6, 7, 8, 13, 29

LQ Median UQ

Find the interquartile range:

$Q_1 = 4, \quad Q_3 = 7.5$

$IQR = Q_3 - Q_1 = 7.5 - 4 = 3.5$

Find the upper and lower bound for the outliers:

Lower bound $= Q_1 - 1.5\,(IQR)$

$4 - 1.5\,(3.5) = -1.25$ So there are no outliers on the lower tail.

Upper bound $= Q_3 - 1.5\,(IQR)$

$7.5 + 1.5\,(3.5) = 12.75$ So there are two outliers on the upper tail.

Outliers are 13 and 29

(ii) The number 29 is clearly an error because this is not a child.

The number 13 may be an error but is unlikely. We should investigate further before removing it from the data set.

Remove the number 29 only

Copyright © Save My Exams. All Rights Reserved

Examiner Tips and Tricks

- Read the question carefully to determine which type of outlier you should be finding and to make sure you are using the correct method.

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.